

GEORGE JOBI PERANGATTU

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PROFESSIONAL SUMMARY

Robotics Software Engineer pursuing an M.S. in Robotics and Autonomous Systems (AI) at Arizona State University, with a B.S. in Computer Science. Hands-on experience developing autonomous robotics software, real-hardware localization and control, field-style validation workflows, and sensor fusion systems using C/C++, Python, ROS/ROS 2, Linux, LiDAR, cameras, and depth sensors. Experienced with debugging, troubleshooting, field tests, data analysis, motion planning, SLAM fundamentals, Git-based collaboration, safety checks, and technical documentation for autonomous systems.

EDUCATION

M.S. Robotics and Autonomous Systems (AI Concentration) Graduating Jul 2026

Arizona State University, Tempe, AZ

B.S. Computer Science Graduated May 2025

Arizona State University, Tempe, AZ

TECHNICAL SKILLS

Role Emphasis: Field Robotics, Autonomous Vehicles, Autonomous Driving Stack, ROS2, Linux, Git, Field Tests, Diagnostics, Real-Time Troubleshooting, Sensor Integration, Shell Scripting Fundamentals, Linear Algebra, Geometry, Probability

Localization & SLAM: Robot Localization, Localization Algorithms, SLAM Fundamentals, State Estimation, Kalman Filter Fundamentals, Particle Filter Fundamentals, Sensor Fusion, Camera-LiDAR Fusion, Pose Estimation, Depth Estimation, Mapping, Navigation, Positioning

Robotic Sensor Systems: LiDAR, Stereo Cameras, RGB Cameras, RGB-D Cameras, Depth Sensors, Point Cloud Processing, Object Detection, Segmentation, BEV Perception, 3D Perception, Computer Vision, OpenCV, Sensor Integration

Autonomous Robotics Software: ROS/ROS 2, Gazebo Simulation, CARLA, URDF, Motion Planning (RRT), Path Planning, Trajectory Optimization, Collision Avoidance, Kinematics, PID Control, Closed-Loop Control, Autonomous Systems

Languages & Platforms: C/C++ (Modern), Python, C, Java, JavaScript, TypeScript, SQL, Linux-Based Platforms, Linux/Ubuntu, Embedded Platforms Fundamentals ML & Deep Learning: PyTorch, TensorFlow, Transformers, CNNs, Multi-Task Learning, Mixed-Precision Training, Machine Learning, Data-Driven Analysis, NumPy, Pandas

Software Engineering: Git, CMake, CI/CD, Docker, Code Review, Agile, Testing, Validation, Production Code, Documentation, Problem-Solving, GDB, Algorithms, Data Structures

LOCALIZATION & PERCEPTION PROJECT EXPERIENCE

Efficient TransFuser: Camera-LiDAR Sensor Fusion for Localization & Perception Spring 2026

Arizona State University - CSE 598 Research Project

- Developed and optimized robot localization and perception algorithms by designing a camera-LiDAR sensor fusion pipeline, replacing TransFuser's backbone with EfficientNetV2-S to reduce parameters by 70% (168M to 50M) while improving held-out loss by 8.7%.
- Built multi-scale transformer fusion for RGB image and LiDAR point cloud processing, achieving 25.4% BEV segmentation improvement for object detection, pose estimation, depth estimation, and localization in the CARLA autonomous driving simulator.
- Integrated multi-task optimization across five perception tasks (localization-relevant depth estimation, mapping, semantic segmentation, detection, and waypoint prediction), using data-driven analysis to validate system performance across all perception outputs.
- Diagnosed an AMP instability in CenterNet's focal loss under FP16, documenting, validating, and maintaining high-quality production code with rigorous testing and problem-solving methodology.

Parrot MiniDrone - Real-Time Localization & Autonomous Control Spring 2026

Arizona State University - Robotics & Autonomous Systems Coursework

- Emphasized physical robot testing, troubleshooting, sensor logging, and real-time debugging for field robotics environments.
- Developed robot localization algorithms on a physical robotic platform, implementing closed-loop PID control (yaw, velocity, altitude) with sensor fusion of inertial and visual sensor data for real-time state estimation and positioning on Linux-based embedded hardware.
- Built automated test sequences to validate localization and control performance using data-driven analysis, logging sensor data from robotic sensor systems and measuring system functionality through iterative performance metrics.
- Engineered safety checks and failsafes ensuring reliable autonomous operation under real-time constraints, demonstrating strong problem-solving skills on real robotic hardware.

3D Motion Planning for Autonomous Navigation in Gazebo Fall 2025

Arizona State University - Robotics & Autonomous Systems Coursework

- Developed autonomous robotics software (3D RRT) in Python for collision-free trajectory computation, integrated with Gazebo simulation using ROS for testing, validation, and data-driven analysis of navigation and localization-relevant path planning performance.
- Applied path post-processing to optimize trajectory quality for downstream robotic control, researching and implementing new algorithms and approaches for improved autonomous navigation.

Spider CAD Robot - Multi-Legged Robot Design (In Progress) Spring 2026

Arizona State University - Robotics & Autonomous Systems Coursework

- Designing a multi-legged robot in CAD with constrained joints, validating kinematics and actuator placement; preparing a URDF-exportable model for localization and controls experimentation in ROS/Gazebo simulation.

Factory Automation System for Semiconductor Process Simulation Starting Jun 2026

Arizona State University - Applied Project (Upcoming)

- Will develop an industrial automation system integrating robotic sensor systems, actuators, and basic PLC control logic for a manufacturing process simulation, applying sensor integration and localization-relevant state tracking across hardware interfaces.

WORK EXPERIENCE

Backend Developer Intern - DigiClips Media Aug 2024 - May 2025

Tempe, AZ

- Optimized data processing pipelines (10M+ daily queries) using Python and ML, reducing latency by 45% - using data-driven analysis to validate system performance and functionality improvements.
- Developed computer vision algorithms for image recognition with cross-functional teams, improving perception accuracy by 30% through testing, validation, and iterative problem-solving.

Full Stack Developer Intern - Odoo May 2025 - Aug 2025

Dubai, UAE

- Refactored Python backend for production applications, documenting, validating, and maintaining high-quality production code in collaboration with multidisciplinary engineering teams.

Full Stack Developer Intern - TicketDex Jan 2025 - May 2025

Tempe, AZ

- Delivered core backend (Node.js) with testing and validation; refactored database layer achieving 30% latency reduction using data-driven analysis and strong communication and collaboration skills.